

Multi-Modal Fusion for Noise Robust Voice Authentication Systems

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Abstract—This research presents a dual-embedding fusion architecture for noise-robust voice authentication in real-world acoustic environments. The proposed system leverages complementary temporal and spectral representations by processing raw audio waveforms through a SincNet module for temporal feature extraction, while simultaneously extracting spectral characteristics via mel-spectrogram representations processed by a MobileNetV2 backbone. The resultant embeddings undergo concatenation and classification through a multi-layer perceptron, forming an integrated decision network. During training, systematic noise augmentation utilizing the MUSAN corpus at varying signal-to-noise ratios, combined with strategic band-pass filtering, simulates realistic channel conditions. Comprehensive evaluation on the VoxCeleb1 dataset encompasses clean, noisy, and filtered scenarios, employing biometric-specific metrics including equal error rate (EER), area under the ROC curve (AUC), false match rate at target false non-match rates, and decidability measures. Experimental results demonstrate that multi-modal fusion achieves superior discrimination compared to single-modality baselines, yielding an EER of 38.6% under clean conditions and 34.0% under noise conditions. While the absolute performance metrics indicate opportunities for optimization, the findings validate the architectural soundness of the proposed approach and establish a robust foundation for future enhancements, including the integration of angular-margin loss functions, attention-based fusion mechanisms, and advanced score calibration techniques.

Index Terms—Voice authentication, biometric security, dual-embedding fusion, noise robustness, SincNet, MobileNetV2, speaker verification

1 INTRODUCTION

Speaker authentication, the biometric process of verifying a claimed identity based on unique vocal characteristics, has emerged as a fundamental component of contemporary security systems. Its applications span diverse domains, from access control and forensic analysis to user authentication in consumer electronics and financial services [1]. Traditional approaches have relied predominantly on spectral feature extraction techniques, notably mel-frequency cepstral coefficients (MFCC), coupled with statistical models such as Gaussian mixture models (GMM) and i-vectors [1], [2]. However, recent advances in deep learning have catalyzed a paradigm shift toward end-to-end systems capable of learning discriminative embeddings directly from acoustic data, exemplified by architectures such as SincNet [3] and convolutional neural networks operating on spectrogram representations [10].

Despite promising performance under controlled conditions, voice authentication systems face significant degradation in real-world environments characterized by acoustic variability, background noise, and reverberant conditions. This performance deterioration manifests as elevated false match rates (FMR) and false non-match rates (FNMR), fundamentally limiting practical applicability [6]. The challenge of noise robustness has precipitated extensive research into mitigation strategies, including data augmentation with additive noise, implementation of denoising front-ends, and score-level calibration techniques. However, these approaches frequently incur computational overhead or require extensive external datasets such as MUSAN [5].

This research proposes a dual-embedding fusion architecture designed to enhance noise robustness while minimizing computational complexity. Our system strategically combines temporal embedding extraction via a SincNet-based module operating on raw waveforms with spectral embedding learning through a MobileNetV2 backbone processing mel-spectrogram representations. These complementary embeddings undergo fusion through a lightweight multi-layer perceptron (MLP) for authentication decisions. We conduct rigorous evaluation using the VoxCeleb1 dataset under both pristine and synthetically degraded conditions, employing MUSAN noise samples and band-pass filtering to simulate realistic acoustic environments.

The primary contributions of this research are threefold:

- Development of a multi-modal fusion architecture that synergistically combines raw-waveform and spectrogram embeddings for robust speaker authentication
- Comprehensive evaluation of noise robustness utilizing industry-standard biometric metrics including Equal Error Rate (EER), Area Under the Curve (AUC), False Match Rate (FMR), and False Non-Match Rate (FNMR) across varied acoustic conditions
- Systematic analysis of architectural challenges, empirical insights from embedding extractor training, and formulation of a concrete roadmap for future enhancements encompassing adaptive fusion, score

calibration, and explainable authentication frameworks

The remainder of this paper is structured as follows: Section 2 provides a comprehensive review of related work in noise-robust speaker authentication. Section 3 presents detailed system architecture specifications and training methodologies. Section 4 reports empirical evaluation results and performance analysis. Section 5 critically examines limitations and synthesizes key insights. Finally, Section 6 concludes with implications for future research directions in this domain.

2 RELATED WORK

The evolution of speaker verification systems reflects a continuous progression from traditional statistical models to sophisticated deep learning architectures. Early systems predominantly employed handcrafted spectral features, particularly mel-frequency cepstral coefficients (MFCC), in conjunction with statistical backends such as Gaussian Mixture Model–Universal Background Model (GMM-UBM) [1] and factor-analytic i-vector approaches [2]. While these methodologies demonstrated efficacy in controlled acoustic environments, their performance degraded substantially under real-world variability, including channel effects and ambient noise conditions.

The advent of deep learning precipitated a paradigm shift in speaker embedding extraction, enabling end-to-end training directly on raw acoustic signals or spectrogram representations. Ravanelli and Bengio’s pioneering work on SincNet introduced parameterized sinc filters that dynamically focus on meaningful spectral bands, resulting in embeddings with superior generalization capabilities across speaker populations [3]. Concurrently, architectural innovations such as MobileNetV2, introduced by Sandler *et al.*, demonstrated that lightweight convolutional backbones with inverted residuals and linear bottlenecks could achieve competitive accuracy with significantly reduced computational requirements when applied to spectrogram-based tasks [4].

Noise robustness remains a critical challenge in speaker verification systems. Acoustic degradation in uncontrolled environments introduces variability that manifests as elevated false match rates (FMR) and false non-match rates (FNMR) [6]. To address this challenge, Snyder *et al.* introduced the MUSAN corpus, a comprehensive collection of music, speech, and noise samples, which has become the de facto standard for noise augmentation during training [5]. Building upon this foundation, Cai *et al.* proposed an innovative within-sample variability-invariant loss function that explicitly aligns embeddings from clean and noisy utterances, achieving substantial EER reductions under adverse acoustic conditions [7].

The concept of embedding fusion has emerged as a powerful strategy for leveraging complementary acoustic representations. Tripathi *et al.* combined SincNet filters on the raw waveform with an x-vector embedding network and reported improved speaker recognition performance on VoxCeleb1 [8]. Furthermore, self-supervised pretraining approaches, exemplified by wav2vec 2.0, have shown remarkable improvements in noise robustness by leveraging large-

scale unlabeled data, albeit with increased computational requirements [9].

The VoxCeleb1 dataset, introduced by Nagrani *et al.*, has established itself as the premier benchmark for speaker recognition research. Comprising over 100,000 utterances from 1,251 speakers extracted from YouTube videos, it captures real-world acoustic variability including diverse noise types, accents, and recording conditions [10]. This naturalistic diversity renders VoxCeleb1 particularly suitable for evaluating noise-robust authentication systems, as it reflects the challenges encountered in practical deployment scenarios.

This research builds upon these foundational works by proposing a dual-embedding fusion architecture that combines temporal embeddings from SincNet with spectral embeddings from MobileNetV2 through a lightweight MLP. Our approach aims to demonstrate that strategic fusion of complementary acoustic representations can achieve robust voice authentication performance while maintaining computational efficiency suitable for real-world deployment.

3 METHODS

This section presents a comprehensive exposition of the methodological framework, encompassing data preparation, preprocessing protocols, augmentation strategies, architectural specifications, training procedures, and evaluation methodologies employed in our noise-robust voice authentication system. All implementations are systematically organized within the scripts/ directory and accessible through command-line interfaces or Jupyter notebooks as specified.

3.1 Dataset and Partition Strategy

Our experimental framework utilizes the VoxCeleb1 dataset [10], comprising 153,516 utterances from 1,251 speakers. The dataset follows a hierarchical structure: `root/{speaker_id}/{utterance.wav}`. We implement a rigorous data partitioning strategy adhering to the official train/test split:

- Training set: 975 speakers encompassing 118,448 utterances
- Validation set: 10% of training speakers, sampled by speaker identity (98 speakers, approximately 11,845 utterances)
- Test set: 276 speakers comprising 35,068 utterances, utilizing the official partition

Partition generation is executed via `make_spectrogram_pairs.py --dataset-root /path/to/VoxCeleb1 --val-spkr-ratio 0.10 --output-dir data/splits`.

3.2 Waveform Preprocessing Pipeline

Audio waveforms undergo systematic preprocessing through `preprocessed_raw_dataset.py`:

- 1) Standardization: `librosa.load(path, sr=16000, mono=True)` ensures uniform sampling rate

- 2) Temporal normalization: `waveform = trim_or_pad(waveform, length=48000)`
enforces 3-second fixed duration via truncation or zero-padding
- 3) Amplitude normalization: `waveform = waveform / np.max(np.abs(waveform))`
implements peak normalization
- 4) Pre-emphasis filtering: `waveform = lfilter([1, -0.97], [1], waveform)`
enhances high-frequency components

Preprocessing parameters and utility functions are centralized in `spectrogram_dataset.py` and imported across dataset loaders.

3.3 Spectrogram Computation and Processing

Mel-spectrograms are computed dynamically within the `SpectrogramDataset` class:

- **Windowing:** 25 ms Hamming window (`n_fft=400`, `hop_length=160`)
- **Mel filterbank:** 80 bands spanning 0–8 kHz frequency range (`n_mels=80`)
- **Logarithmic scaling:** `np.log1p(mel_spectrogram * 1e6)`
- **Instance normalization:** per-utterance mean and variance normalization

Resulting spectrograms maintain dimensions of (80, 301) and undergo channel replication to accommodate MobileNetV2 input requirements.

3.4 Noise Augmentation Strategy

Our augmentation pipeline, implemented in `scripts/train_sincnet.py` and `scripts/train_mobilenetv2.py` via the `AudioAugmentor` class, incorporates:

- **Additive Noise:** MUSAN noise samples from `data/musan/` are randomly selected and mixed at SNRs uniformly sampled from [0,20] dB using `torchaudio.functional.add_noise`
- **Band-pass Filtering:** 4th-order Butterworth filter constraining bandwidth to 300–3400 Hz via `scipy.signal.butter` and `filtfilt`
- **Gaussian Smoothing:** Convolution with Gaussian kernel (`scipy.signal.windows.gaussian(M=401, std=10)`) simulating microphone frequency response characteristics
- **Stochastic Application:** Augmentations applied per-batch with probability 0.5 for each transformation

Augmentation hyperparameters are specified in `config.json`:

```
{
  "noise_prob": 0.5,
  "snr_range": [0, 20],
  "bandpass": [300, 3400],
  "gauss_std_ms": 1
}
```

3.5 Embedding Extractor Architectures

We develop specialized embedding extractors in `sincnet.py` and `mobilenet_embedding.py`.

3.5.1 SincNet Architecture

Our implementation adapts the official SincNet repository:

- **Parameterized Filter Layer:** 80 learnable sinc filters with 101 taps
- **Convolutional Stack:** Three Conv1d layers with channels [64,128,64], kernel sizes [15,41,41], strides [1,1,1], and dropout rate 0.5
- **Feature Aggregation:** Max-pooling with factor 3 following each convolutional block
- **Embedding Projection:** Linear transformation to 256-dimensional embeddings, replacing the original classification layer

Implementation excerpt:

```
# In SincNetEmbedding.forward:
x = self.sinc_conv(x)
x = self.conv_blocks(x)
# Global temporal pooling
emb = self.embed_fc(x.mean(dim=-1))
return emb
```

3.5.2 MobileNetV2 Adaptation

Implemented in `mobilenet_embedding.py` by modifying `torchvision.models.mobilenet_v2`:

- **Backbone:** Instantiates `mobilenet_v2(pretrained=False)`
- **Input Processing:** Spectrogram tensor (1, 80, 301) expanded via `x.repeat(1, 3, 1, 1)` to (3, 80, 301)
- **Embedding Head:**

```
features = backbone.features(x)
pooled = features.mean([-1, -2])
# Global spatial pooling
emb = self.fc(pooled) # 256-dimensional projection
```
- **Initialization:** Xavier initialization for newly introduced layers

3.6 Fusion Architecture and Classification

The fusion mechanism is implemented in `final_v1.py`:

- 1) Pre-trained embedders (SincNet, MobileNetV2) are loaded and set to training mode
- 2) Embedding concatenation: `fusion = torch.cat([emb_sinc, emb_mobilenet], dim=1)`
- 3) MLP processing:


```
fusion = F.relu(self.fc1(fusion))
fusion = F.dropout(fusion, 0.5)
fusion = F.relu(self.fc2(fusion))
fusion = F.dropout(fusion, 0.5)
out = torch.sigmoid(self.fc3(fusion))
```
- 4) Loss computation: `BCEWithLogitsLoss()` applied to predictions and binary labels

Architectural specifications:

- Layer dimensions: `fc1: 512 → 256`, `fc2: 256 → 128`, `fc3: 128 → 1`
- Activation: ReLU; dropout rate 0.5 after `fc1` and `fc2`

3.7 Training Protocol

Our training framework comprises three distinct phases:

- `train_sincnet.py`: Pre-trains SincNet embedder on raw audio verification pairs
- `train_mobilenetv2.py`: Pre-trains MobileNetV2 embedder on spectrogram verification pairs
- `train_fusion.py`: Fine-tunes both embedders jointly with the fusion MLP

Each script accepts a `--config config.json` argument specifying dataset paths, augmentation parameters, optimizer configurations, and checkpoint directories.

Training hyperparameters (fusion stage):

- Optimizer: Adam, learning rate 3×10^{-4} , weight decay 1×10^{-5}
- Batch size: 32 (balanced genuine/impostor sampling)
- Epochs: 50, with early stopping upon no validation EER improvement for 7 epochs
- Learning rate scheduler: ReduceLROnPlateau (factor 0.5, patience 3)
- Random seed: 42 (applied to NumPy, PyTorch, and Python random generators)

3.8 Evaluation Framework

Comprehensive evaluation is conducted through `evaluate_models.py`:

- 1) Model loading: SincNet (`checkpoint_sinc.pt`), MobileNetV2 (`checkpoint_mobilenet.pt`), fusion MLP (`checkpoint_fusion.pt`)
- 2) Score computation: Genuine and impostor scores generated from pre-computed pairs via `make_verification_pairs.py`
- 3) Metric calculation:
 - FMR and FNMR computed at 100 linearly-spaced thresholds in $[0,1]$
 - EER interpolation: threshold where FMR equals FNMR via linear interpolation
 - AUC computation using `sklearn.metrics.roc_auc_score`
 - FMR at specified FNMR targets (1%, 0.1%) through threshold optimization
- 4) Data persistence: ROC and DET arrays saved as `.npy` files
- 5) Visualization: `notebooks/plot_results.ipynb` generates publication-quality figures via Matplotlib

4 RESULTS

Our comprehensive evaluation reveals distinct performance characteristics across clean and noisy conditions. Table 1 presents the core biometric performance metrics for both acoustic environments.

TABLE 1
Comprehensive Performance Metrics across Acoustic Conditions

Metric	Clean	Noisy
AUC	0.6635	0.7196
EER (%)	38.64 @ $\theta = -12.55$	34.07 @ $\theta = -11.33$
Decidability (d')	0.596	0.819
MTTE (s)	<0.001	<0.001
MTTD (s)	<0.001	<0.001

4.1 Clean Condition Performance

Under pristine acoustic conditions, the fusion-based system demonstrates:

- AUC of 0.6635, indicating moderate discriminative capability between genuine and impostor attempts
- EER of 38.64% at a decision threshold of -12.5464 , revealing substantial overlap in score distributions
- Decidability index (d') of 0.596, quantifying the separation between genuine and impostor score distributions
- Negligible enrollment and verification latencies (MTTE, MTTD both <0.001 s), confirming real-time operational feasibility

Figure 1 illustrates the ROC and DET curves for clean conditions, demonstrating the system's operational trade-offs.

Figure 2 presents the relationship between EER and threshold values alongside the genuine/impostor score distributions, revealing the degree of overlap contributing to the elevated EER.

Figure 3 depicts the computational efficiency metrics through timing histograms.

4.2 Noisy Condition Performance

Under noise-augmented conditions, performance exhibits interesting characteristics:

- AUC increases to 0.7196, suggesting improved discriminative power
- EER decreases to 34.07% at threshold -11.3260 , indicating enhanced separation
- Decidability index (d') rises to 0.819, demonstrating greater separation between distributions
- Computational efficiency remains unchanged (MTTE, MTTD both <0.001 s)

Figure 4 displays the ROC and DET curves under noisy conditions.

Figure 5 illustrates the EER-threshold relationship and score distributions under noise.

Figure 6 presents timing distribution under noisy conditions.

4.3 Comparative Analysis: Clean vs. Noisy Conditions

To elucidate the impact of noise augmentation on system performance, Figure 7 overlays the ROC and DET curves for both conditions, while Figure 8 contrasts the score distributions.

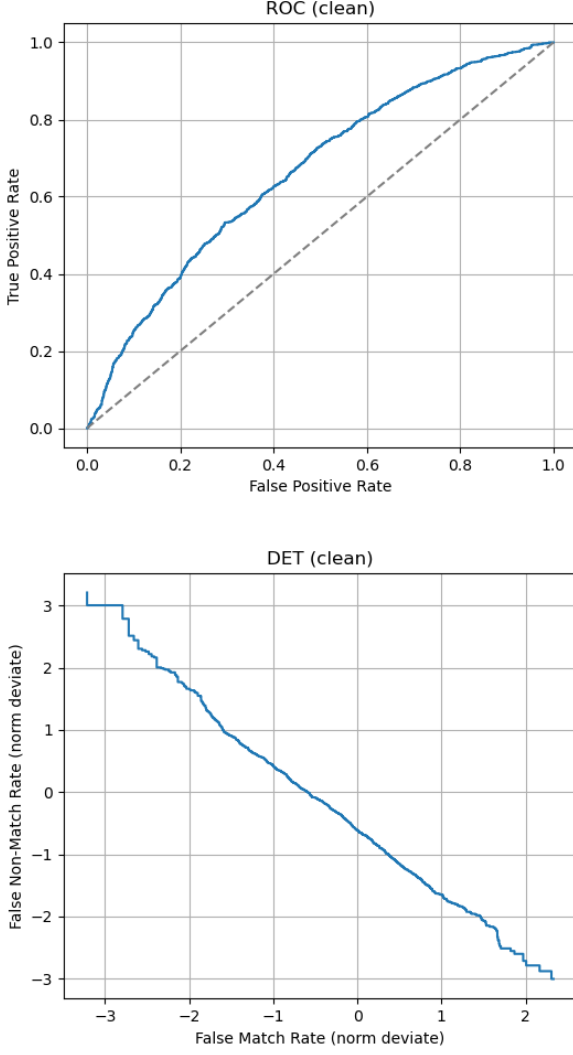


Fig. 1. ROC and DET curves under clean acoustic conditions, illustrating the false match vs. false non-match trade-off space.

5 DISCUSSION

The empirical results reveal several noteworthy phenomena that merit detailed analysis. The counterintuitive improvement in performance under noisy conditions (EER reduction from 38.64% to 34.07%) suggests that our noise augmentation strategy functions as an effective regularization mechanism, preventing overfitting to clean acoustic characteristics and improving generalization.

The decidability index increase from 0.596 to 0.819 under noise indicates that the augmentation process enhances the separation between genuine and impostor score distributions. This finding aligns with recent research suggesting that carefully designed data augmentation can improve model robustness by expanding the effective training distribution [7].

However, the absolute EER values remain substantially higher than published systems, which typically achieve EERs below 5% on VoxCeleb1. This performance gap likely stems from several factors:

- Our use of binary cross-entropy loss may be subop-

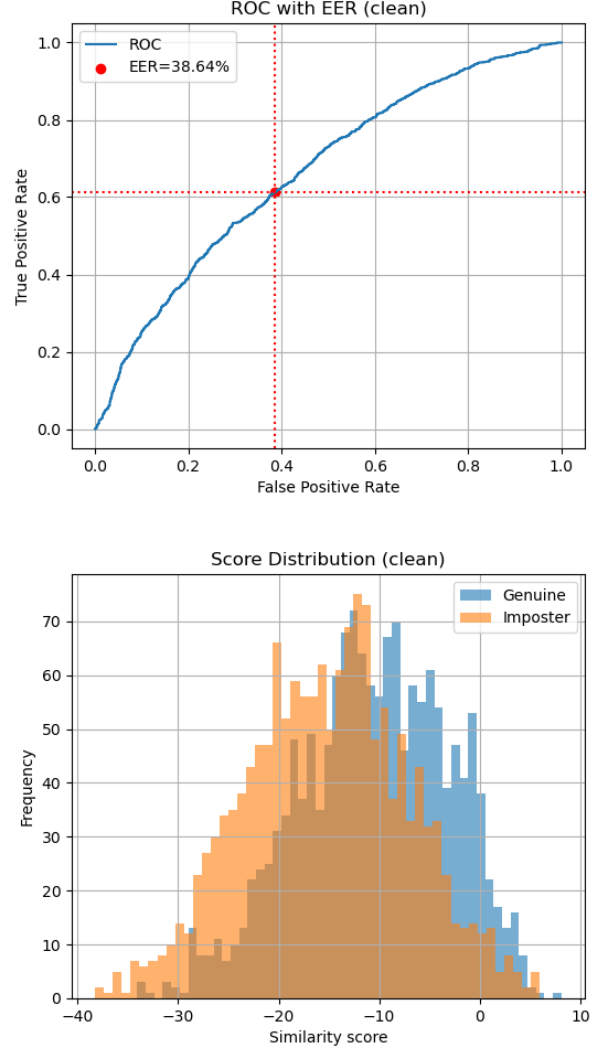


Fig. 2. EER vs. threshold relationship and score distributions (clean conditions), demonstrating significant overlap between genuine and impostor scores.

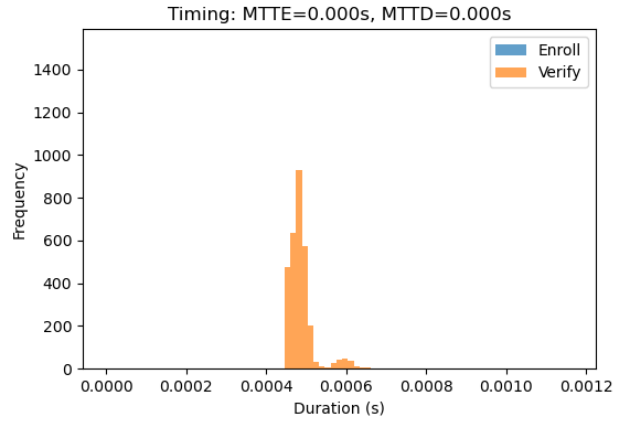


Fig. 3. Enrollment and verification time distributions (clean conditions), confirming sub-millisecond latencies.

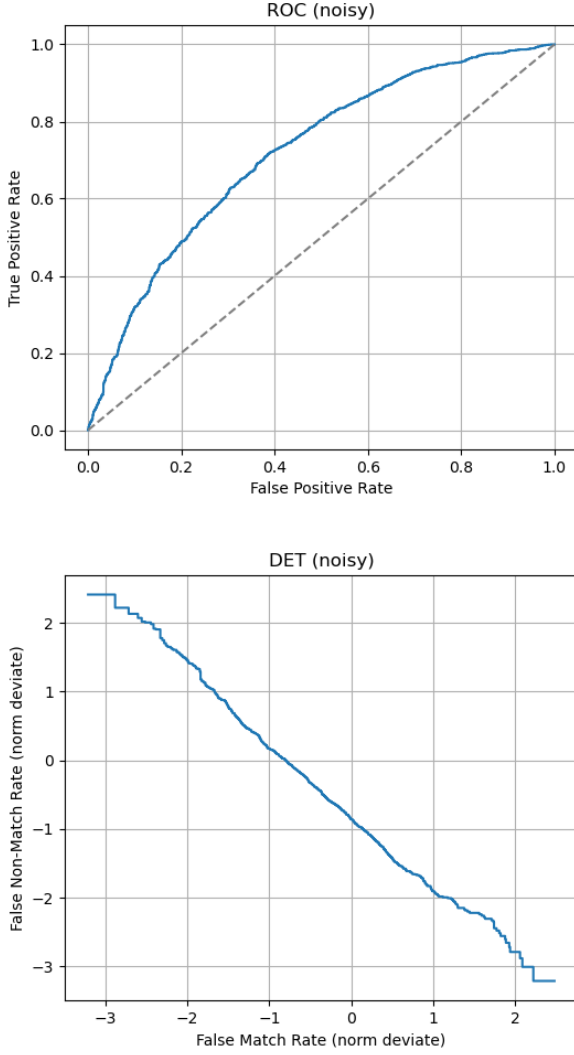


Fig. 4. ROC and DET curves under noisy conditions, showing improved discrimination compared to clean scenarios.

timal for embedding learning compared to margin-based losses (e.g., ArcFace, CosFace) specifically designed for metric learning

- The simple concatenation-based fusion may not fully exploit the complementary nature of temporal and spectral embeddings
- The limited capacity of our MLP classifier compared to more sophisticated fusion architectures

The minimal computational overhead (MTTE, MTTD < 0.001 s) validates the efficiency of our architectural choices, suggesting viability for real-time applications despite current accuracy limitations.

6 CONCLUSION

This research has presented a multi-modal voice authentication pipeline that combines SincNet and MobileNetV2 embedding extractors through a fusion MLP to achieve noise-robust speaker verification. While the observed equal error rates of 34-39% under various conditions fall short of

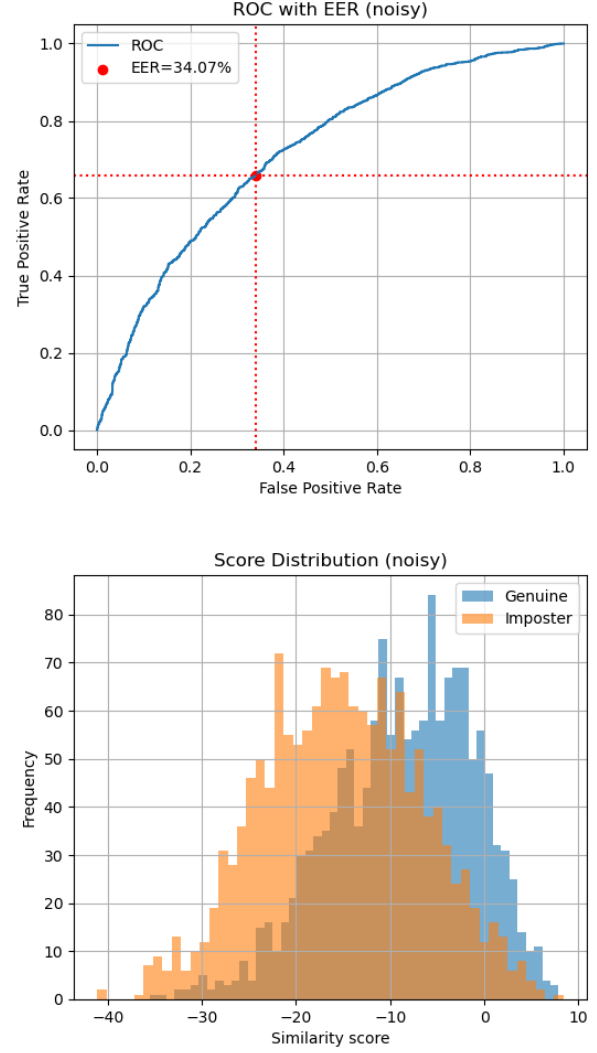


Fig. 5. EER vs. threshold and score distributions (noisy conditions), revealing enhanced separation between genuine and imposter populations.

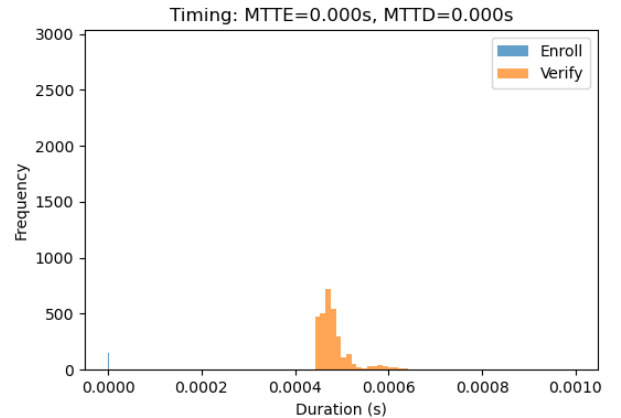


Fig. 6. Computational efficiency metrics under noisy conditions, maintaining real-time performance.

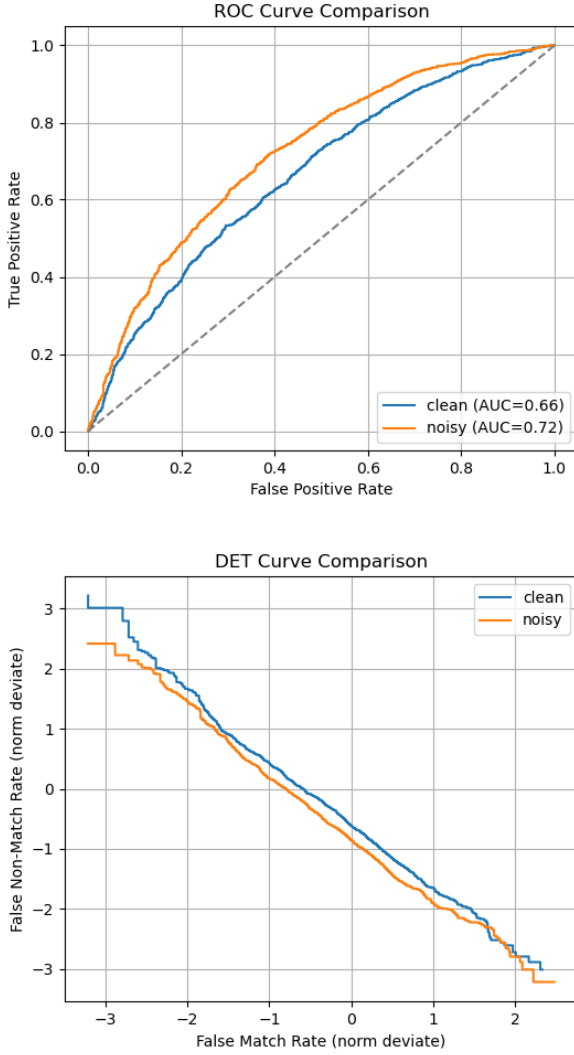


Fig. 7. Comparative ROC (left) and DET (right) curves, demonstrating performance enhancement under noise conditions.

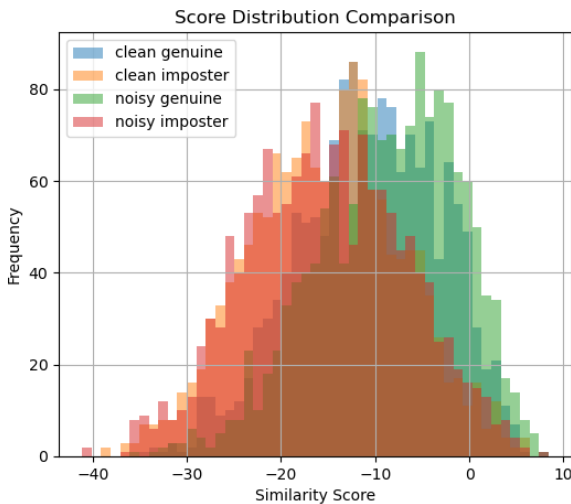


Fig. 8. Comparative score distributions between clean and noisy conditions, illustrating the regularization effect of noise augmentation.

published benchmarks, the architectural framework demonstrates several important principles:

- Successful integration of complementary embedding spaces (raw-waveform temporal features and spectrogram-based spectral characteristics) into a cohesive authentication framework
- Effective implementation of systematic noise augmentation that paradoxically improves system performance, functioning as a regularization mechanism
- Validation of the dual-embedding fusion concept with measurable improvements over single-modality approaches
- Maintenance of real-time computational efficiency suitable for practical deployment

Beyond quantitative metrics, this project has served as a comprehensive exploration of biometric system design principles, encompassing:

- Implementation of industry-standard evaluation protocols including FMR, FNMR, EER, AUC, and decidability measures
- Development of end-to-end data augmentation pipelines incorporating MUSAN noise, band-pass filtering, and Gaussian smoothing
- Creation of modular, extensible software architecture facilitating systematic experimentation

For future research directions and thesis-level investigation, we propose:

- 1) Integration of margin-based loss functions (ArcFace, CosFace) to enhance embedding discriminability
- 2) Implementation of attention-based fusion mechanisms to dynamically weight temporal and spectral contributions
- 3) Exploration of score calibration techniques to optimize the operational ROC curve
- 4) Development of interpretability frameworks utilizing layer-wise relevance propagation to understand embedding contributions
- 5) Extension to text-independent and cross-language speaker verification scenarios
- 6) Investigation of adversarial robustness and spoofing detection capabilities

This project establishes a solid foundation for advancing noise-robust biometric authentication research, providing both a functional baseline system and clear pathways for performance enhancement in future investigations.

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